# Supplementary material

qPCRtools: a cross-platform desktop application for reproducible qPCR analysis without programming

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## 1 Analysis output on the bundled example data

Figure S1 shows the two analyses discussed in the main text, produced from the application's numerical output by the plotting script shipped in the repository (`code/validation/plot_examples.py`, matplotlib), which gives finer control over the layout than the in-interface figure export. The per-group means, standard deviations and $P$-values were independently reproduced without the application or the R package by `code/validation/check_example_stats.R`.

## 2 Reproducible re-analysis of published qPCR data

The two real example plates distributed with the 2022 *qPCRtools* paper were processed independently in the graphical interface. The four-gene dilution series was used to demonstrate curve fitting and quality-control rejection. The separate treatment plate contains six genes, two groups, four biological replicates and two technical replicates; its design table supplies the published gene-specific efficiencies. Because those efficiencies are not all equal to the reference gene, the efficiency-corrected standard-curve expression path was selected. The treatment data and dilution series are separate published examples and are not described as one biological experiment.

## 3 Validation against an independent base-R reference

The application and an independent base-R implementation were run on the same input files for all three quantification methods. The reference was computed directly from the raw $C_q$ tables and did not call the R package. Every resulting quantity was compared pairwise (Figure S2, Table S1).

The maximum relative difference over all compared values was $1.9 \times 10^{-14}$. Numerical agreement establishes consistency with the specified algorithms, not biological correctness.

For the ANOVA path, the overall ANOVA $P$ value was 0.0034471 and the Tukey compact letter display was b / ab / a. Group 0 and group 1 differed ($P_{\text{adj}} = 0.0028$), whereas group 0.5 was not separated from either ($P_{\text{adj}} = 0.052$ and 0.073). The Holm-adjusted Welch comparison for the 0.5 group was $P = 0.0121$, so its symbol changed from ** to * relative to the unadjusted value.

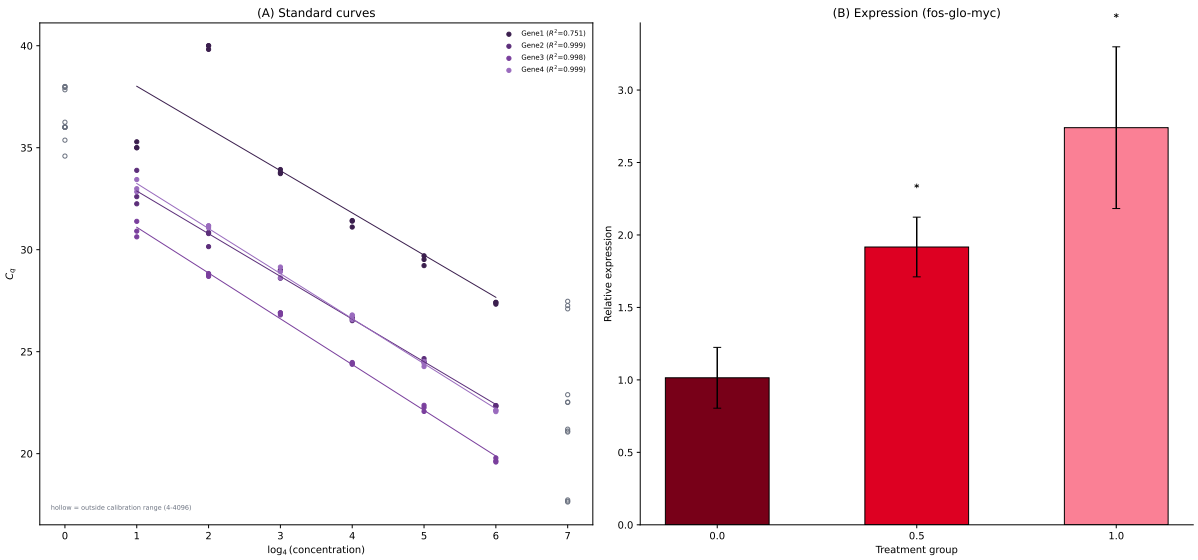



Figure S1: Analysis output on real data. (A) Standard curves for four genes over a four-fold dilution series, with fitted regressions and $R^2$. The abscissa is $\log_4$ concentration (dilution steps), matching the regression. Hollow symbols mark wells outside the calibrated range (4–4096). Gene1 has $R^2 = 0.75$ and is unsuitable for quantification; the other three have $R^2 > 0.99$. (B) Relative expression of *fos-glo-myc* across treatment groups ($\Delta\Delta C_t$ method, calibrated to the control group), with Holm-adjusted significance labels and individual biological replicates. Reproduced from `figures/fig2_results_v2.pdf`.

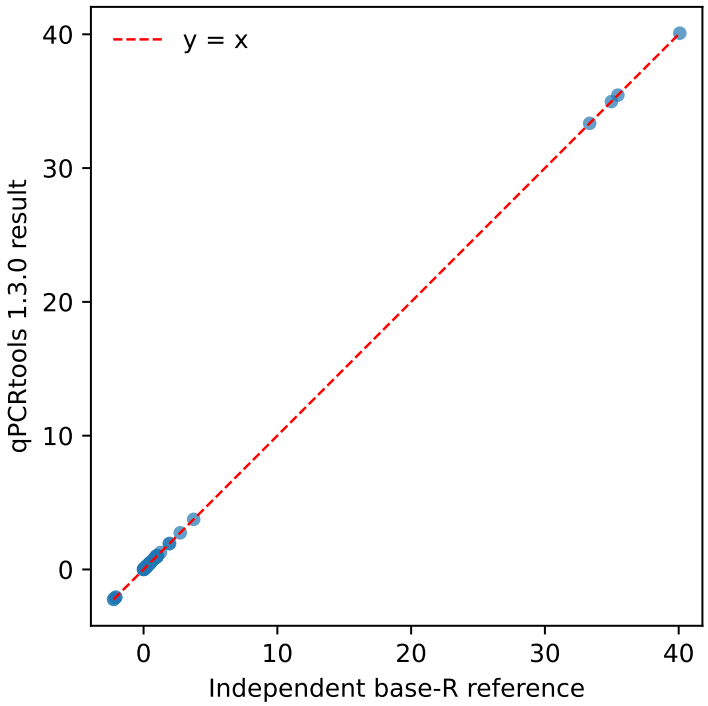



Figure S2: Agreement between the application and an independent base-R reference. Each application result is plotted against the corresponding reference value across all methods; all points fall on the line $y = x$.

Table S1: Agreement of the application with an independent base-R reference. (a) Compared quantities across all three methods; the maximum relative difference over all 58 paired values was $1.9 \times 10^{-14}$. (b) Selected values from the bundled $\Delta\Delta C_t$ example. The unadjusted $P$ values (0.006027 and 0.022191) are retained, and Holm adjustment within the gene gives 0.012055 and 0.022191. Generated by `code/validation/reference_base_r.R` and `code/validation/compare_to_reference.R`.

*(a) agreement with the independent base-R reference*

| Method | Compared quantities | Max. rel. diff. |
|---|---|---|
| $\Delta\Delta C_t$ | expression, SD, raw and Holm $P$ | $1.7 \times 10^{-14}$ |
| Standard curve | slope, intercept, $R^2$, efficiency | $4.2 \times 10^{-15}$ |
| RqPCR | expression, SD, SE, raw and Holm $P$ | $1.9 \times 10^{-14}$ |

*(b) selected values (bundled example, $\Delta\Delta C_t$)*

| Quantity | Application (C++) | Independent base-R reference |
|---|---|---|
| Mean ± SD, group 0.0 | $1.0145 \pm 0.2098$ | $1.0145 \pm 0.2098$ |
| Mean ± SD, group 0.5 | $1.9169 \pm 0.2060$ | $1.9169 \pm 0.2060$ |
| Mean ± SD, group 1.0 | $2.7403 \pm 0.5580$ | $2.7403 \pm 0.5580$ |
| Welch raw $P$ vs control | 0.006027 / 0.022191 | 0.006027 / 0.022191 |
| Welch Holm $P$ vs control | 0.012055 / 0.022191 | 0.012055 / 0.022191 |
| ANOVA $P$ (all groups) | 0.0034471 | 0.0034471 |
| Tukey letters | b / ab / a | b / ab / a |

Table S2: Sensitivity of the Welch comparison to the analysis scale. The primary analysis uses linear relative expression; the alternative uses $\Delta C_t$. Holm correction is applied across the two treatment-control comparisons on each scale.

| Scale and group | Raw $P$ | Holm-adjusted $P$ |
|---|---|---|
| Linear relative expression, dose 0.5 | 0.006027 | 0.012055 |
| Linear relative expression, dose 1 | 0.022191 | 0.022191 |
| $\Delta C_t$, dose 0.5 | 0.017565 | 0.017565 |
| $\Delta C_t$, dose 1 | 0.004404 | 0.008809 |

Table S3: End-to-end re-analysis of the published qPCRtools examples. The two plates are separate examples from the 2022 study. The workflow was run through the graphical interface and checked by `code/scripts/19_verify_real_workflow.py`.

| Stage | Input and settings | Result |
|---|---|---|
| Dilution series | 96 wells; four genes; eight four-fold concentrations (4–4096); three wells per point | Gene1 $R^2 = 0.7509$ and was rejected; the other curves had $R^2 =$ 0.9983–0.9993 |
| Method selection | Published gene-specific efficiencies for six genes | Standard-curve expression selected instead of $2^{-\Delta\Delta C_t}$ |
| Group comparison | 96 wells; five target genes plus *OsUBQ*; CK versus treatment; four biological and two technical replicates | Welch tests with Holm correction: *OsPR10* $P = 0.0368$; *OsWAK91* $P = 0.00230$; the other three genes were not significant |
| QC and export | Default Cq and technical-replicate thresholds; summary, replicate-level table, two figures and analysis record exported | One well with $C_q > 35$; 13 samples with technical-replicate SD > 0.5; deterministic checker returned `REAL-WORKFLOW PASS` |